Mrinal Mishra, Ph.D.

Bioinformatics Analyst. Computational genomics and machine learning for pharmacogenomics.

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My career on a time axis, drawn like a genome browser. The interactive version is on the website. Genomics Machine learning Software

Bioinformatics Analyst and computational biologist with a Ph.D. in Biology and hands-on experience building genomic analysis workflows, machine learning models, and multi-omics studies. Focused on pharmacogenomics, gene expression prediction, variant interpretation, population genomics, and research that improves representation of historically excluded populations in precision medicine.

Current work

Perera Lab BioPortal

Full-stack web app for running genomics tools from a browser

- Web platform for authenticated lab users to upload genomics files, queue command-line bioinformatics jobs, monitor progress, and download outputs.
- Built with Django REST Framework, React, Celery, Redis, PostgreSQL, and Docker. Supports chunked uploads, file validation, background jobs, audit logging, and output bundles.
- Tool catalog includes PLINK 1.9/2.0, GCTA, Beagle, FLARE, GWAS/TWAS, variant-calling, RNA-seq, and QC workflows.

ALA_GEM

Local-ancestry-aware gene expression modeling

- Training and validation workflows around genotype, local ancestry, expression, and covariate data on hg38.
- Chromosome-level runs, nested cross-validation, elastic-net style modeling, and external validation comparisons.

Ancestry-aware bioinformatics AI system

Confidential project, described at a high level only

- A guided AI analysis layer for variant, region, population, association, enrichment, and quality-control questions, with emphasis on scope control, reproducibility, and explainable summaries.

Experience

Bioinformatics Analyst

2022 to present

Northwestern University, Feinberg School of Medicine. Perera Lab, Department of Pharmacology

Develop machine learning and genomics workflows for pharmacogenomics research, with emphasis on African American cohorts and gene expression regulation.

- Building deep learning models to predict gene expression across the human genome.
- Analyzing drug treatment effects in hepatic cells from African American samples.
- Contributed to peer-reviewed research on drug perturbation, eQTL mapping, and multi-omics equity in precision medicine.

Ph.D. Candidate, Computational Biology

2015 to 2021

University of Florida, Department of Biology

Executed whole-genome sequencing analysis, functional annotation, gene enrichment, variant detection, and population genomics research.

- Researched hybridization of four-eyed sleeper fish populations across regions of the East China Sea.
- Identified, evaluated, and categorized population variants for dissertation and publication work.
- Created R workflows for population genomics and developed a Python package to filter low-quality variants.

Graduate Teaching Assistant

2015 to 2021

University of Florida, Department of Biology

- Mentored students in individual and group settings on coursework and project work.
- Designed and delivered bioinformatics workshops for undergraduate students.
- Taught DNA/RNA structure, gene expression, proteomics, PCR, gel electrophoresis, and fermentation.

Research Trainee

2012 to 2014

VTT Technical Research Centre, Turku, Finland

Applied statistical modeling and image analysis to study cellular heterogeneity in 3D cancer tumor-cell images.

- Developed a cell image analysis pipeline covering acquisition, quality control, cleanup, exploratory analysis, and hypothesis-driven analysis.
- Wrote R scripts to segment cancer cell images and extract morphological feature sets.
- Prepared raw data for statistical analysis and machine learning models.

Summer Research Trainee

2010

Vellore Institute of Technology, Tamil Nadu, India

- Conducted in silico pathway analysis of pancreatic cancer genes, focusing on KRAS, CDKN2A, MADH4, TP53, and ARMET.
- Integrated expression-profile data into molecular interaction networks using Cytoscape and presented the findings as a poster.

Publications

Mishra M*, Nahlawi L*, Zhong Y, De T, Yang G, Alarcon C, Perera MA. LA-GEM: imputation of gene expression with incorporation of Local Ancestry. *Pac Symp Biocomput.* 2024:341. PMID: 38160291. *Equal contribution.

Yang G, **Mishra M**, Perera MA. Multi-Omics Studies in Historically Excluded Populations: The Road to Equity. *Clin Pharmacol Ther.* 2023;113(3):541-556. doi: 10.1002/cpt.2818.

Zhong Y, De T, **Mishra M**, Avitia J, Alarcon C, Perera MA. Leveraging drug perturbation to reveal genetic regulators of hepatic gene expression in African Americans. *Am J Hum Genet.* 2023;110(1):58-70. doi: 10.1016/j.ajhg.2022.12.005.

Liu Q, **Mishra M**, Saxena AS, Wu H, Qiu Y, Zhang X, You X, Ding S, Miyamoto MM. Balancing selection maintains ancient polymorphisms at conserved enhancers for olfactory receptor genes of a Chinese marine fish. *Molecular Ecology.* 2021;30(16):4023-4038.

Li H, **Mishra M**, Ding S, Miyamoto MM. Diversity and Dynamics of *Candidatus* Endobugula and Other Symbiotic Bacteria in Chinese Populations of the Bryozoan, *Bugula neritina*. *Microbial Ecology.* 2019;77(1):243-256.

Ding S, **Mishra M**, Wu H, Liang S, Miyamoto MM. Characterization of hybridization within a secondary contact region of *Bostrychus sinensis* in the East China Sea. *Heredity.* 2018;120(1):51-62.

Mishra M, Liang S, Ke C, Ding S, Miyamoto MM. The Potential Utility of Microsatellite Markers for Resolving the Phylogeographic Structure of the Rock Shell in the Northwest Pacific. *J Phylogenetics Evol Biol.* 2016;4:4.

Gupta R, **Mishra M**, Tandon P, Hisaria R. Novel mutation predicted to disrupt SGOL1 protein function. *Egyptian Journal of Medical Human Genetics.* 2013;14:149-155.

Mishra M, Kumar A. Computational analysis of genetic network involved in pancreatic cancer in human. *BMC Bioinformatics.* 2011;12 Suppl 11:A11.

Technical stack

Programming	R, Python, Biopython
Machine learning	TensorFlow, Keras, PyTorch, scikit-learn, regression, convolutional neural networks, autoencoders
GWAS and sequencing	GATK, BWA, Bowtie, DESeq2, BayeScan, KAT, variant calling, eQTL mapping
Annotation	MAKER, Blast2GO
Population genetics	mothur, FSTAT, Genepop, STRUCTURE, GenAlEx, POPGENE, GDA, Micro-Checker, MrBayes, TNT, MEGA
Software engineering	Django REST Framework, React, Celery, Redis, PostgreSQL, Docker, Nginx

Education

Ph.D. in Biology, 2021
University of Florida, Gainesville, FL

M.S. in Bioinformatics, 2015
University of Turku, Turku, Finland

B.Tech in Bioinformatics, 2012
Vellore Institute of Technology, Vellore, India

Awards and service

Grinter Fellowship, 2015
University of Florida

Travel Fellowship Award, 2013
Swiss Institute of Bioinformatics

Best Poster Award, 2011
International Society for Computational Biology

Conference volunteer
ISMB/ECCB Vienna 2011 and Berlin 2013

Certifications

Supervised Machine Learning: Regression and Classification

Advanced Learning Algorithms

Python for Genomic Data Science

Genomic Data Science with Galaxy

Introduction to Genomic Technologies

Languages

English, full professional proficiency

Hindi, native

Finnish and German, elementary